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Child acquisition of complex consonants:

The role of oral motor control in developing acoustic targets

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Our study examines the role of oral motor control in the acquisition of acoustic targets for lingually complex consonants in Estonian by speakers from four age groups. We focus on children's production of secondarily palatalised and non-palatalised consonants where anatomical development and acquisition of motor control skills make differing predictions about the timing and nature of speech production. Our aim is to disentangle the contribution of, on the one hand, speech motor control development and, on the other hand, anatomical growth and articulator change in proportion. We examined oral motor control using a single syllable alternating motion rate diadochokinetic (DDK) task and studied the difference between F2−F1 using Functional Principal Component Analysis. Our results show that palatalised and non-palatalised productions differ significantly across all age groups indicating children can produce palatalisation contrasts from age 3 years, but consistency in production increases with age. Results also show that children and adults with greater oral motor control skills are better able to produce the palatalisation contrast. These results suggest that anatomical proportions pre-dispose children towards early acquisition of the acoustic properties of palatalisation. However, adult-like production of the phonological contrast also depends on the acquisition of oral motor control skills.

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I. INTRODUCTION

In this paper, we examine the role of oral motor control in the acquisition of acoustic targets for lingually complex consonants. Child speech acquisition involves learning the cognitive categories for language, but also the physical ability to produce them (Kent, 1992). During this process, a child must learn the skills necessary for precise movements of the speech articulators and adapt these movements responsively as their anatomy develops and changes in proportion. We focus on child production of consonants /l, t, n, s/ which can also be phonemically palatalised /l^{j}, t^{j}, n^{j}, s^{j}/ in Estonian. Research on anatomical development and acquisition of speech motor control skills make differing predictions about the timing and nature of speech production. Large cross-linguistic comparisons indicate that consonants with multiple lingual articulations are typically acquired later on in the developmental trajectory (McLeod and Crowe, 2018). This is hypothesised to be due to the development of speech motor control skills needed for precise tongue placement. However, in secondary palatalisation, a child's anatomical proportions might predispose them to produce palatal-sounding speech (Ciusa *et al.*, 2007; Vihman and Vihman, 2011).

Our aim is to disentangle the contribution of, on the one hand, motor control development and, on the other hand, anatomical growth and articulator change in proportion. In doing so, we test long-standing hypotheses about the order of acquisition for complex consonants in child speech. Additionally, we diversify child speech studies in two respects. First, we provide a detailed account of the acoustic acquisition of secondary palatalisation. Second, we respond to calls in the literature for more child language data from non-Indo-European

languages (Kidd and Garcia, 2022) by analysing data from Estonian, a Finno-Ugric language, and providing normative data on oral motor control development.

In the remainder of this section, we discuss previous phonetic research which has considered the trajectory of consonant acquisition and the role of motor control skills (Section I A). Second, we discuss anatomical development and how this might predispose children to palatalise (Section I B). Third, we discuss secondary palatalisation in Estonian (Section I C), before outlining our research questions and hypotheses (Section I D).

A. Acquisition of oral motor control and complex consonants

It is generally observed that children acquire consonants in a hierarchical fashion of increasing difficulty and complexity (Shriberg, 1993). In addition to increasing complexity, the individual child's abilities and the environment in which a child grows up also play a role (Davis and Bedore, 2013). Our analysis focusses on the articulatory and physical factors in child speech development. We assess the role of individual differences in the acquisition of speech motor control, and how this relates to consonant production at different ages. Cross-linguistic surveys have been instrumental in recognising consonant acquisition trends across the world and understanding the role of articulatory factors in general patterns of child acquisition. McLeod and Crowe (2018) assess normative data from 27 languages to identify that consonants with one primary articulation tend to be acquired earlier. For example /p, m, t/ are acquired by 2;11 with up to 100% accuracy. Consonants with multiple articulations or those requiring precise tongue placement and timing are usually acquired later, for example, /k^{w}, t^{ʕ}/ are typically acquired by 4;11 (Table 1 in McLeod and Crowe,

2018, 1551). Previous detailed acoustic and articulatory work also indicates that segments requiring multiple lingual gestures are among the last sounds to be acquired, for example, coda lateral /l/ (Lin and Demuth, 2015), and rhotic /r/ (Klein *et al.*, 2013).

When children learn to produce speech sounds, they have to rely on somatosensory and auditory feedback to map and correlate articulatory movements to specific speech sounds considering the current context of the system (Guenther, 2006). As children's anatomy is constantly changing, they must adjust, refine and correct their sensory mappings based on the auditory (Ayala *et al.*, 2023) and articulatory feedback (Tremblay *et al.*, 2003). This feedback loop is important for the speech to be fluent (Guenther, 2006). With age, auditory feedback becomes less important and sensorimotor representations reliably predict the motor commands needed to produce speech sounds (Bishai *et al.*, 2026). Our focus is on consonants requiring multiple lingual articulations. To produce such sounds, children need to gain control in particular of the superior longitudinal muscle through auditory and somatosensory feedback for intelligible production(Abakarova *et al.*, 2022, 3295).

To gain a window into the development of speech motor control and how this leads eventually to adult-like production, researchers have pursued two main strategies: first, investigating the overlap of segments through coarticulation, and secondly, by investigating tongue flexibility and timing. Coarticulatory research has revealed differences between adults and children, and suggests that child coarticulation patterns do not become adult-like until adolescence (Zharkova, 2018). This supports earlier work using other methods such as EMA (electromagnetic articulography), which indicates that motor control over the speech articulators does not stabilise until late adolescence (Cheng *et al.*, 2007b). In general, the

research programme in child coarticulation finds that children coarticulate more than adults (Abakarova *et al.*, 2022; Noiray *et al.*, 2018; Rubertus and Noiray, 2018, 2024), though others find that children coarticulate less than adults (Barbier *et al.*, 2020; Zharkova, 2018). Child articulation is also more variable than adult articulation (Barbier *et al.*, 2020).

This body of research in child speech articulation shows clear differences in adult and child timing and consistency of gestures. One of the reasons cited for these differences is maturation of the fine motor control needed to differentiate and coordinate segments in an adult-like fashion (Abakarova *et al.*, 2022; Barbier *et al.*, 2020). Specifically, acquisition of the ability to manipulate the different parts of the tongue is referred to as 'lingual differentiation' (Gibbon, 1999). Lingual differentiation is essential for the production of consonants with multiple lingual articulations. Articulatory work on lingual differentiation indicates that differentiation increases with age, and that children with Speech Sound Disorder might have difficulties in acquiring differentiation (Cleland and Scobbie, 2021; Dokovova *et al.*, 2023; Kabakoff *et al.*, 2021). Crucially, acquisition of lingual differentiation must be accompanied by acquisition of adult-like timing patterns in order for differentiated gestures to be perceived as appropriately coarticulated adult-like speech (Cychosz, 2021; Howson and Redford, 2021; Kpogo and Chang, 2025).

In order to measure the acquisition of oral motor control, we used a diadochokinetic speech task (DDK), a tool commonly used by Speech and Language Therapists and researchers to assess the extent of articulator agility, mobility and oral motor coordination (Duffy, 2013; Kent *et al.*, 2022). In those DDK tasks participants are asked to clearly and rapidly repeat a variety of syllables, syllable combinations or words in a given amount of time. Although

researchers have questioned the usefulness of studying non-speech oral movements as it uses different mechanism to meaningful speech (Ziegler *et al.*, 2019), the method is still widely used in clinical assessment, multilingualism, and child research (e.g. Chu *et al.*, 2021a; Ha, 2023; Ong *et al.*, 2024). In child research concentrating on typically developing children, the findings indicate that overall the repetition rate is increasing with age (Ha, 2023; Murali *et al.*, 2024).

Specifically in the production of monosyllables, children are slower than adults and three-year olds slower than older children (Ha, 2023). Although DDK rate increases with age, the relationship is not linear. Children from 3–5 have been shown to have similar rates but increase in accuracy and consistency during that time (Stackhouse, 2000). DDK rate in adults has been shown to slow down with age and older people are slower than middle-aged with no apparent gender differences in that trend (Chu *et al.*, 2021b). Interestingly, because of phonetic and phonological differences between languages, DDK rates also vary cross-linguistically and the rates are not always directly comparable between languages (Chu *et al.*, 2021a). Estonian Speech and Language Therapists have reported that they regularly use DDK tasks in their work to help to diagnose issues with oro-motor control (Lahtein-Kürsa *et al.*, 2025). However, despite the widespread use of this tool, there are currently no normative data of DDK scores at different ages in Estonian. An additional contribution of our study is to provide a first set of data for this purpose, noting that fuller results will follow from Lahtein-Kürsa *et al.* (in preparation).

We assess the role of oral motor control in the acquisition of acoustic targets for consonants requiring multiple lingual articulations. We assume that lingual differentiation and timing

will be essential in this process, and we expect to observe developmental differences. Our case study involves the context of secondary palatalisation, which challenges some of the previously held assumptions about motor control and speech acquisition, as we discuss in the next sub-section.

B. Acquiring secondary palatalisation

We define 'complex consonants' as those requiring multiple independently controlled articulatory constrictions (Shaw *et al.*, 2021, 438). Shaw et al.'s work indicates that complex segments, e.g. /pʲ/, differ from segment sequences, e.g. /pj/, in their coordination patterns and gestural landmarks (Gafos, 2002). Within the grouping of complex consonants, we focus on phonemic secondary palatalisation. For these consonants in coda position (as in Estonian, Section I C), it is argued that phonemic palatalisation is timed in-phase with consonantal gestures, though timing varies with individual and contextual perceptual maximisation (Bennett *et al.*, 2025).

In secondary palatalisation, a consonant has a primary place of articulation, e.g., the alveolar ridge, and also a secondary constriction at the hard palate (Bateman, 2011). This additional palatalisation gesture can form the basis of distinguishing minimal pairs, e.g. *kann* /kɑnː/ 'jug' and *kann* /kɑnʲː/ 'toy' in Estonian. The literature discussed above would predict that secondary palatalisation should be acquired later on in child development due to the multiple lingual constrictions required. However, there are indications that secondarily palatalised consonants might be acquired earlier than other complex segments. For example, in Russian, a language with an extensive system of secondary palatalisation contrasts,

children sometimes acquire palatalised /tʲ/ before /t/ (Zharkova, 2005, 191). Similarly, Timm (1977)'s diary study of Russian acquisition by one child notes that /gʲ/ and /kʲ/ are produced with fewer errors than /g/ and /k/. Russian child speech has long been noted as characterised by an 'overall softness' (palatalised quality), which is attributed to the evolution of the vocal tract morphology (Zharkova, 2005, 195).

Zharkova provides an observational and acoustic analysis of the acquisition of the palatalisation contrast in seven Russian L1 children aged 1;3 to 3;2. She notes that production is quite variable and children appear to be testing different strategies until age 3 years. The oldest child in the sample produces speech which perceptually approximates adult targets. In a diary study of acquisition of Estonian and American English bilingual child from 12–21 months, researchers found that she frequently produced forms with a 'palatal template' where her first words were produced with high front vowels, or front rising diphthongs (Vihman and Vihman, 2011, 113). This showed that the child used palatal forms at a higher rate than the input received from her parents (Vihman and Vihman, 2011, 119).

Existing studies of the acquisition of Russian and Estonian suggest anatomical reasons behind the early acquisition of secondary palatalisation and the overall palatal quality of child speech (Vihman, 1971; Vihman and Vihman, 2011; Zharkova, 2005). When compared to three-year old children, six-year olds have greater palatal height, greater palatal slope, and an overall more domed palate shape (Ciusa *et al.*, 2007). Similarly, EPG data indicate greater palatal contact in children aged six years compared to adolescents and adults (Cheng *et al.*, 2007a), and also more fronted tongue articulations in six-year olds compared to adolescents (Fletcher, 1989). In the development of the vocal tract, many of the speech

articulators undergo a period of very rapid evolution around 13–14 months and begin to take on adult-like proportions, for example, vocal tract length, tongue length, naso-oro-pharyngeal length, and laryngeal height. The hard palate, however, continues to evolve very rapidly until approximately 25 months (Vorperian *et al.*, 2005, 342). This means that while the child is producing their first words and longer utterances, the ratio of tongue size to hard palate length is not in adult-like proportion. As such, they either need to make articulatory adaptations to achieve adult-like acoustic targets, or they will produce speech with a palatal quality.

To summarise: two contrasting factors might contribute to the acquisition of palatalisation in child speech: 1) the process of acquiring tongue motor control and timing patterns (predicting difficulty in acquiring lingually complex consonants), and 2) anatomical factors such as palate shape and the relative length of the hard palate in younger children (predicting earlier and increased palatalisation). Our study investigates the relationship between the acquisition of motor control and acoustic targets for phonemic secondary palatalisation. If children who have greater motor control skills have more adult-like production of the palatalisation contrast, then production of these complex consonants is related to the development of speech motor control. If there is no or a weaker relationship between tongue control and palatalisation, then production of the palatalisation contrast is more closely related to anatomical development than acquisition of tongue motor control.

C. Secondary palatalisation in Estonian

Estonian coronal consonants /l, t, n, s/ can be non-palatalised or palatalised /lʲ, tʲ, nʲ, sʲ/ (Asu and Teras, 2009). Palatalised consonants can only occur in syllable codas, either at the end of monosyllabic words, or after the first stressed syllable in disyllabic words (Teras and Pajusalu, 2014). Typologically, Estonian is unusual in only having a palatalisation contrast in coda position. Typically, languages which have phonemic secondary palatalisation have it in onsets and then variably in codas. For example, contrastive palatalisation in codas has been neutralised in Bulgarian (Iskarous and Kavitskaya, 2018) and in Nenets (Kochetov, 2002).

Estonian Palatalisation results in minimal pairs e.g., *kott* /kotː/ 'worn out shoe' and *kott* /kotʲː/ 'bag' and is not shown in contemporary orthography. For Estonian palatalised consonants, the primary place of articulation on the alveolar ridge is accompanied by a near-simultaneous constriction with the tongue body at the hard palate (Malmi and Lippus, 2019). Estonian palatalised consonants are prepalatalised, meaning that most of the perceptual and acoustic quality of palatalisation is carried by the preceding vowel. This leads to a front-raising diphthongal quality to vowels preceding palatalised consonants and increased duration (Malmi *et al.*, 2023; Teras and Pajusalu, 2014). The importance of the preceding vowel was highlighted in a perception experiment: Estonian listeners were not able to accurately identify palatalised consonants when the transition preceding a palatalised consonant was removed (Liiv, 1965).

Dynamic acoustic analysis of palatalised and non-palatalised vowels and consonants in adult speech is presented in Malmi *et al.* (2023). This study indicates that in adult speech, F2 values during the vowel preceding palatalisation reliably distinguish palatalised and non-palatalised consonants at all places of articulation and in all vowel contexts. In all cases, F2 is higher in palatalisation. F1 tends to be lower in palatalisation, and F3 does not systematically vary according to palatalisation. Articulatorily those changes in the formant values reflect the higher and more fronted tongue position with palatalisation (Malmi and Lippus, 2019). These findings confirm other descriptions of secondary palatalisation cross-linguistically, where most often raised F2 and lowered F1 are reported in the vowels surrounding palatalised consonants (Kochetov, 2017; Nance and Kirkham, 2020). In Estonian, the timepoint where formant trajectories in vowels preceding palatalised consonants diverge from those preceding non-palatalised consonants is at vowel onset or in the first 50% of vowel duration for the majority of possible contexts and consonants (Malmi *et al.*, 2023). In contrast, acoustic cues for palatalisation in Estonian consonants themselves are less consistent and typically affect only the first part of the consonant (Malmi *et al.*, 2023, 769). Together, these findings suggest that adult-like palatalisation in Estonian can robustly be identified through analysis of F1 and F2 in the vowels preceding palatalised consonants.

D. Research questions and hypotheses

Our study investigates the acquisition of complex consonants by children and adults focussing on secondary palatalisation. We examine the contrasting hypotheses that palatalisation should be acquired later on due to the slower acquisition of fine motor control skills

and manipulation of different tongue muscles, and that palatalisation should be acquired early on due to anatomical biases in child physiology. We include measures of oral motor control ability to test for individual differences in this respect, and provide a first set that we know of normative DDK data for Estonian. We address the following research questions:

1. Are there age-related differences in oral motor control skills in Estonian?
2. Are there age-related differences in the acoustic realisation of secondary palatalisation?
3. Is there a relationship between the development of motor control skills and the acquisition of secondary palatalisation acoustics?

We hypothesise that oral motor control skills increase with age (Ha, 2023; Murali *et al.*, 2024). This has been demonstrated with DDK data in other languages but we need to establish this result first as a basis for Research Questions 2 and 3. As children get older, their ability to coordinate the timing and movement of different articulators improves. We predict that there are developmental differences in the acoustics of palatalisation. Specifically, younger children might not differentiate minimal pairs to the extent that adults do (Malmi and Nance, 2025). We predict that anatomical biases might lead to greater palatalisation in younger children (Ciusa *et al.*, 2007; Zharkova, 2005). We hypothesise that children with greater motor control skills will be able to better manipulate the muscles of the tongue (Dokovova *et al.*, 2023; Gibbon, 1999) and thus differentiate minimal pairs for palatalisation to a greater extent. Due to the expected timing differences in adult and child speech (Howson and Redford, 2021; Noiray *et al.*, 2018), our analysis is designed to incorporate dynamic information about formant trajectories across the vowels preceding palatalised consonants.

II. MATERIALS AND METHODS

A. Participants

The data were collected in three Estonian preschools and schools in the autumn of 2025 and early winter 2026. We worked closely with speech language therapists to find and record typically developing monolingual Estonian children from three age groups: 3;2–4;6 (8F, 10M), 5;6–7;1 (5F, 13M), and 10;0–11;0 (12F, 12M). Additionally, we recorded a comparison group consisting of 20 adults who are first-language Estonian speakers (18–35 years old, 10M, 10F). In total, we analysed data from 80 speakers in this study.

B. Recordings and materials

The children were recorded in a quiet, separate room at their preschools and schools during their school day. Adults were recorded in the phonetics lab at the University of Tartu. With the most cooperative participants, the experiment took around 30 minutes to complete. Once we had obtained the consent forms from the parents or adults, the participants were asked to complete four different tasks. 1) Syllable production (DDK), 2) picture description, 3) perception of palatalisation, and 4) production of individual words. For this study, we focus on tasks 1) and 4). In the first three tasks only audio were recorded using Gorilla (Anwyl-Irvine *et al.*, 2021). In the fourth task both the ultrasound and audio signals were recorded using Articulate Assistant Advanced (AAA) v221.5.11 (Wrench, 2024). All throughout the experiment, the children and adults wore a DPA 4080 Cardioid Lavalier Microphone. The acoustic data recorded in AAA were recorded at a sampling rate of 22,050

Hz (16-bit). To record the ultrasound data, the participants wore a lightweight headset (Spreafico *et al.*, 2018) that securely held the ultrasound probe under the speaker's chin. The current study is part of a larger project that examines the articulatory properties of Estonian children's productions. Here we are looking only at the acoustic properties of their speech.

In our version of the DDK task, we asked each participant to rapidly and accurately repeat the syllable /pɑ/ as many times as possible within a 5-second time window. This was then followed with a trial with /tɑ/, and then /kɑ/. The time window was kept to 5 seconds to be comfortably within the maximum expected phonation time for younger children (Mendes Tavares *et al.*, 2012) and greatly exceeded their maximum capacity that they could produce in one breath. For practice, the participants were first asked to produce different syllables slowly to get the acquainted with the task. We then recorded them producing their best-effort attempt. A full DDK protocol can include many syllable combinations as well as non-speech movements such as lip opening and closing (Kent *et al.*, 2022). We reduced the DDK task to be mindful of the time spent with the children and avoid overburdening them.

The production experiment in AAA consisted of 28 words in isolation including eight minimal pairs for the palatalisation contrast (16 words total) and four one-syllable CV**ː**C containing the vowels /ɑ, e, u, i/ respectively for the purpose of vowel normalisation. We chose common and high frequency words, which were tested in our pilot study (Malmi and Nance, 2025). Also, most of the targets were disyllabic and in nominative case (excl. three) which made them structurally simple. As palatalisation is not represented in contemporary Estonian orthography, and some of the children were not fully literate, we elicited each word

by asking participants to listen and repeat an audio recording while looking at a picture representing the target item similar to Lin and Demuth (2015) (see code repository for our pictures and recordings). It would not be possible to represent all words with pictures alone and without other cues. The audio recordings were made by an L1 Estonian-speaking 37-year old male speaker at the phonetics lab in the University of Tartu using an AKG C414 microphone and Presonus StudioLive 16R soundcard. For adults each test word was repeated five times, for the 10;0–11;0 age group four times, and for the 3;2–4;6 and 5;6–7;1 age groups three times. Here, we analyse the eight minimal pairs for palatalisation, and we used the four CV**ː**C words for the purposes of acoustic normalisation. This word list is shown in Table I.

TABLE I. Word list used in the experiment.

Word	**IPA**	**English**	**Part of speech**	**Category**
kas	/kɑsː/	*whether*	INT.PRON	Minimal pair
kass	/kɑsʲː/	*cat*	SG.NOM	Minimal pair
kus	/kusː/	*where*	INT.PRON	Minimal pair
kuss	/kusʲː/	*shush*	2SG.IMP	Minimal pair
pats	/pɑts/	*pat*	SG.NOM	Minimal pair
pats	/pɑtʲː/	*braid*	SG.NOM	Minimal pair
nutt	/nutː/	*cry*	SG.NOM	Minimal pair
nutt	/nutʲː/	*smarts*	SG.NOM	Minimal pair
kann	/kɑnː/	*jug*	SG.NOM	Minimal pair
kann	/kɑnʲː/	*toy*	SG.NOM	Minimal pair
tunde	/tunːte/	*feeling*	SG.PART	Minimal pair
tunde	/tunʲːte/	*hours*	PL.PART	Minimal pair
tall	/tɑlː/	*lamb*	SG.NOM	Minimal pair
tall	/tɑlʲː/	*barn*	SG.NOM	Minimal pair
mul	/mulː/	*me*	1SG.GEN	Minimal pair
mull	/mulʲː/	*bubble*	SG.NOM	Minimal pair
kaal	/kɑːːl/	*weight*	SG.NOM	Normalisation
keel	/keːːl/	*tongue*	SG.NOM	Normalisation
kuul	/kuːːl/	*moon*	SG.ADD	Normalisation
kiil	/kiːːl/	*dragonfly*	SG.NOM	Normalisation

The youngest children sometimes produced audible palatalisation in words in which the consonant was not phonologically palatalised. As this could be considered a high-stress situation for them, the experimenter asked the children to repeat the word to ensure children had heard it correctly and knew which target word they were aiming for. In most cases, they did not change their pronunciation, and we accepted their second attempt as their

production of that word. As the software used for the recording ultrasound data does not start recording before a beep which occurs approximately a second before pressing record button, the youngest children very regularly imitated the heard stimuli immediately after hearing it, so it was very common to ask them to repeat it again in the same recording.

C. Data processing

We first extracted the syllable repetition recordings from Gorilla and the word list recordings from AAA. The recordings from Gorilla were in WEBM format, which we converted to MP3 using FFmpeg to be readable in Praat (Boersma and Weenik, 2025). The word list recordings were force-aligned using the method in Alumäe *et al.* (2018). We then manually checked the force-aligned boundaries in Praat. Eleven time-normalised F1 and F2 measurement points of the vowel that preceded each consonant were extracted using a script that finds a unique formant ceiling for each of the vowels (see OSF for script). As discussed in Section I C, the acoustics of the vowel preceding each consonant is a robust measure of palatalisation in Estonian (Malmi *et al.*, 2023). For adult male speakers, the maximum ceiling was set at 5000 Hz, for female speakers at 5500 Hz. As we did not expect any sex differences for Estonian children under the age of 12 years (Meister and Meister, 2021), following the Praat manual, the maximum ceiling for both boys and girls was set at 8000 Hz. Further data analysis was carried out using R and RStudio (R Core Team, 2025). In order to compare the formant values of adults and children of differing ages, the data were normalised using the Lobanov method.

D. Oral motor control analysis

To analyse the DDK data, we manually counted the syllables in Praat. We chose this approach over an automated method in order to provide a qualitative judgment as to what counted as a repeated syllable and what did not. We looked for clear breaks in the pitch and intensity traces of the speaker and excluded repetitions in which the separate productions were coarticulated with each other (see 'Supplementary figures' in code repository). We then removed the silences from the recording that was strictly limited to 5 seconds and calculated how many syllables per second were produced by dividing the repetition rates from the time they spent producing syllables (Gadesmann and Miller, 2008).

For the statistical analysis of DDK rates, we used the `lme4` package to fit a linear mixed model where the dependent variable was syllables per second. The independent variables were Age group, Place of Articulation (PoA) (/pɑ/, /tɑ/, /kɑ/), and the interaction between them. Age group and PoA were sum coded. Speaker was included as a random intercept. Following the scoping review in (Kent *et al.*, 2022), typically sex does not affect DDK rates or are not reported in most of the literature. As such, we have pooled data from male and female speakers together in this analysis. In addition, this strategy makes the best use of our statistical power for modelling. For significance testing, we compared the full model with a nested model not containing the predictor of interest via Likelihood Ratio testing in ANOVA (Winter, 2020).

E. Acoustics of palatalisation analysis

In order to capture the time-varying dynamics of the formant movements preceding phonological palatalisation, we first calculated the difference between F2 and F1 (Kochetov, 2017; Nance and Kirkham, 2020). Then, using F2−F1, we fitted a Functional Principal Component Analysis to those values at the 11 timepoints from the start of the vowel to the start of the consonant. Principal Component Analysis (PCA) allows dimensionality reduction of complex data into a smaller number of uncorrelated variables (linear functions), which can then be used to generate scores for each token and subject to further analysis (e.g. Jolliffe and Cadima, 2016). *Functional* Principal Component Analysis (FPCA) allows inclusion of time-series data into a PCA by taking into account that subsequent timepoints are related to one another (Coretta and Sakr, 2026). As the FPCA output is a list of numbers representing formants over time, this can be submitted to analysis by linear mixed effects models allowing us to fit a relatively complex structure which would be less feasible and less interpretable than with other methods of functional data analysis such as Generalised Additive Mixed Models. This method has been most extensively applied to f0 contours (Gubian *et al.*, 2015), but we now apply it to the trajectory of F2−F1.

Following Coretta and Sakr (2026), we fitted an FPCA to the distance between F2−F1 trajectories using the `fda`, `funData`, `MFPCA`, and `landmarkregUtils` packages (Gubian, 2025; Happ and Greven, 2018; Ramsay, 2003). In our data, FPC1 captures 74% of the variation in the data, FPC2 captures only 16%, and the subsequent components capture less and less. Due to the very high proportion of variation captured by FPC1 and big drop-

367 off in variation explained between FPC1 and FPC2, we focus on analysis of the scores for
368 FPC1 only (see Turton, 2017 for a similar strategy with 'standard' PCA). After that, we
369 looked at the interquartile range and removed 90 outlier FPC1 values that exceeded ±3 SD
370 from the mean. We were left with 4460 tokens for the analysis. In order to check that FPC1
371 captures the variation in F2−F1 trajectory as intended, Figure 1 shows the mean of the
372 measured values, and also the mean measured values plus or minus fractions of the standard
373 deviation of FPC1. The figure indicates a rising formant trajectory, which is expected for
374 vowels preceding a combination of palatalised and non-palatalised consonants. When we
375 initially plotted figures of the FPC scores, we observed that FPC1 correlates with *lowness* of
376 F2–F1 i.e. *higher* FPC1 scores meant *lower* F2–F1. For ease of interpretation, we multiplied
377 the FPC1 scores by −1 such that *higher* FPC1 scores can be interpreted as *greater* distance
378 between F2 and F1 (i.e. more palatalisation).

379 The FPC1 scores for the difference between F2−F1 trajectories were modelled as the
380 dependent variable in a linear mixed model. All the discrete variables were sum coded
381 (Age group, Vowel, Palatalisation), and DDK was standardised. As DDK varied by Place
382 of Articulation for all ages (Table II), we used the mean from three different productions
383 as a representative value of their overall performance. Speaker and Word were included as
384 random intercepts. The FPC1 model included two-way interactions for Palatalisation and
385 Vowel, and Palatalisation and Age group. We also included a three-way interaction for Age
386 group, DDK and Palatalisation to test for differing relationships between oral motor control
387 and age in the realisation of the palatalisation contrast.

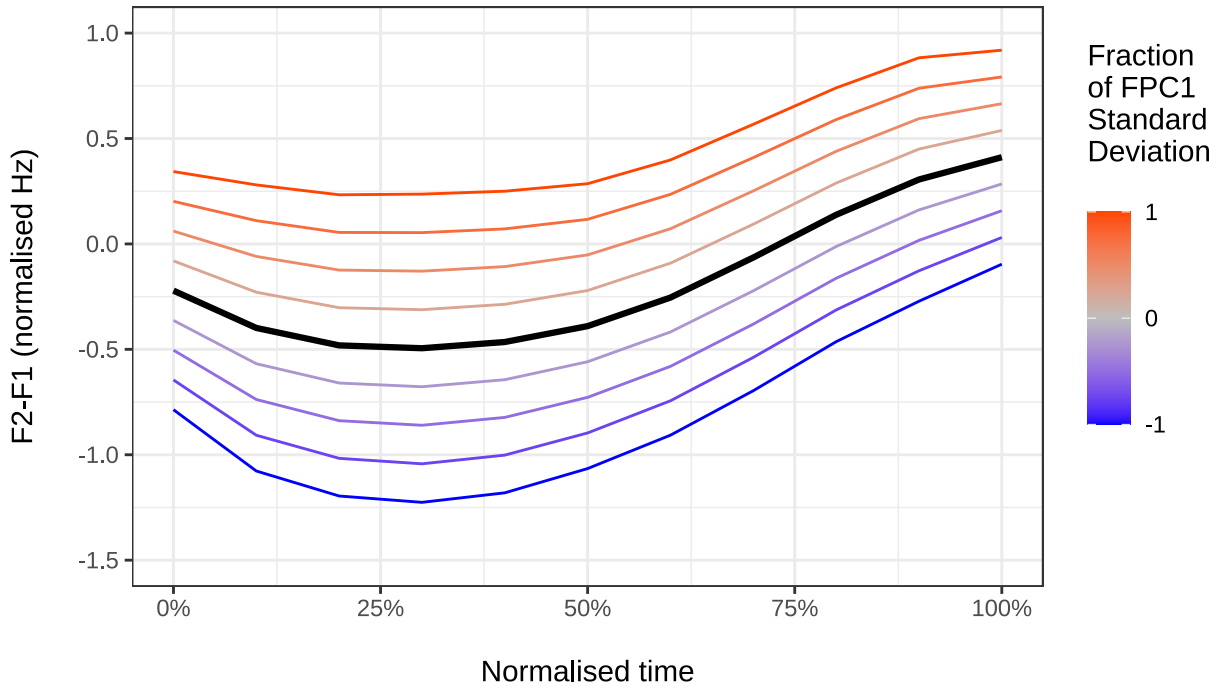


FIG. 1. Interpretation of FPC1 scores. The thick black line shows the mean measured F2. Red and blue lines are the mean F2 ± the Standard Deviation of FPC1.

Where two-way interactions between predictors were significant, we carried out post-hoc testing using the `emmeans` package using the Tukey method and effects plots using the `effects` and `broom.mixed` packages. The significant three-way interaction is interpreted via an effects plot. Our FPC1 analysis does not include sex because the data are normalised with the aim of making adult and child data comparable, and also normalising for physical differences between participants. In adult Estonian, there are almost no differences according to social performance of gender in the realisation of palatalisation contrasts (Malmi *et al.*, 2023, 767), so there is little reason to expect a gender difference among children. Additionally, we wished to avoid excessive model complexity where possible.

III. RESULTS

A. Oral motor control

The raw DDK values of syllables per second by Age group and Place of Articulation (PoA) are plotted in Figure 2, the DDK modelling is presented in Table II, pairwise comparison between Places of Articulation (PoA) is presented in Table IV and comparison between Age groups is in Table III.

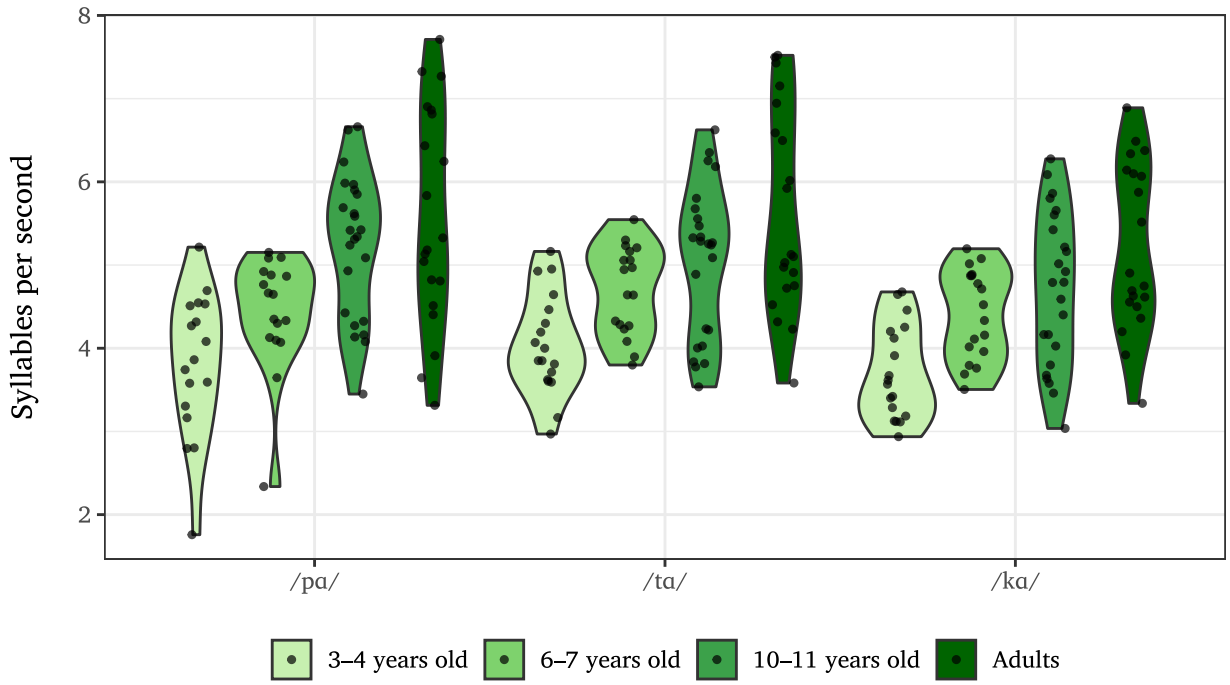


FIG. 2. DDK by Place of Articulation and Age group.

Our analysis showed that Age group is a significant predictor of the rate at which the participants could produce repetitions. The PoA was significant as a main effect but not in the interaction with Age group indicating differences for PoA at all ages.

TABLE II. DDK modelling.

Full model intercept	$\hat{\beta}$	SE($\hat{\beta}$)	t	$p(t)$
	-1	0.19	-5.04	<.001
Fixed effects	*df*		χ^2	$p(\chi^2)$
Age group	9		43.86	<.001
PoA	8		24.78	.002
Age group : PoA	6		6.37	.38

406 The youngest speakers (3–4 years) produced similar number of syllables per second (3.6–
407 5.2, mean 4.2, SD 0.5) to 6–7 year olds (3.8–5.6, mean 4.8, SD 0.5) but significantly fewer
408 than 10–11 year olds (3.5–6.7, mean 5.4, SD 0.8). The youngest speakers and 6–7 year olds
409 both produced significantly fewer syllables per second than adults (3.6–7.7, mean 6, SD 1.2)
410 (Table III). Overall the repetition rates were slower with /kɑ/ than /pɑ/ or /tɑ/ (Table
411 IV).

TABLE III. DDK pairwise comparisons for Age groups.

Age group	$\hat{\beta}$	SE($\hat{\beta}$)	t	p
3–4 years - 6–7 years	-0.61	0.24	-2.54	0.062
3–4 years - 10–11 years	-1.1	0.23	-4.72	<.001
3–4 years - Adults	-1.52	0.24	-6.45	<.001
6–7 years - 10–11 years	-0.45	0.26	-2.01	0.195
6–7 years - Adults	-0.90	0.24	-3.84	0.001
10–11 years - Adults	-0.45	0.22	-2.10	0.176

TABLE IV. DDK pairwise comparisons for Place of Articulation.

PoA	$\hat{\beta}$	SE($\hat{\beta}$)	t	p
/kɑ/ - /pɑ/	-0.26	0.08	-3.10	<.001
/kɑ/ - /tɑ/	-0.33	0.08	-4.11	<.001
/pɑ/ - /tɑ/	-0.11	0.08	-0.92	.630

B. Acoustics of palatalisation

The FPC1 modelling is presented in Table V, pairwise comparisons for Palatalisation and Age group are in Table VI and the pairwise comparisons of Palatalisation, Age group and DDK are in Table VII. Figure 3 shows the effect of Vowel on FPC1 values, Figure 4 shows the model estimated values of FPC1 by Age and Palatalisation, Figure 5 shows the three-way interaction between Age, Palatalisation and DDK.

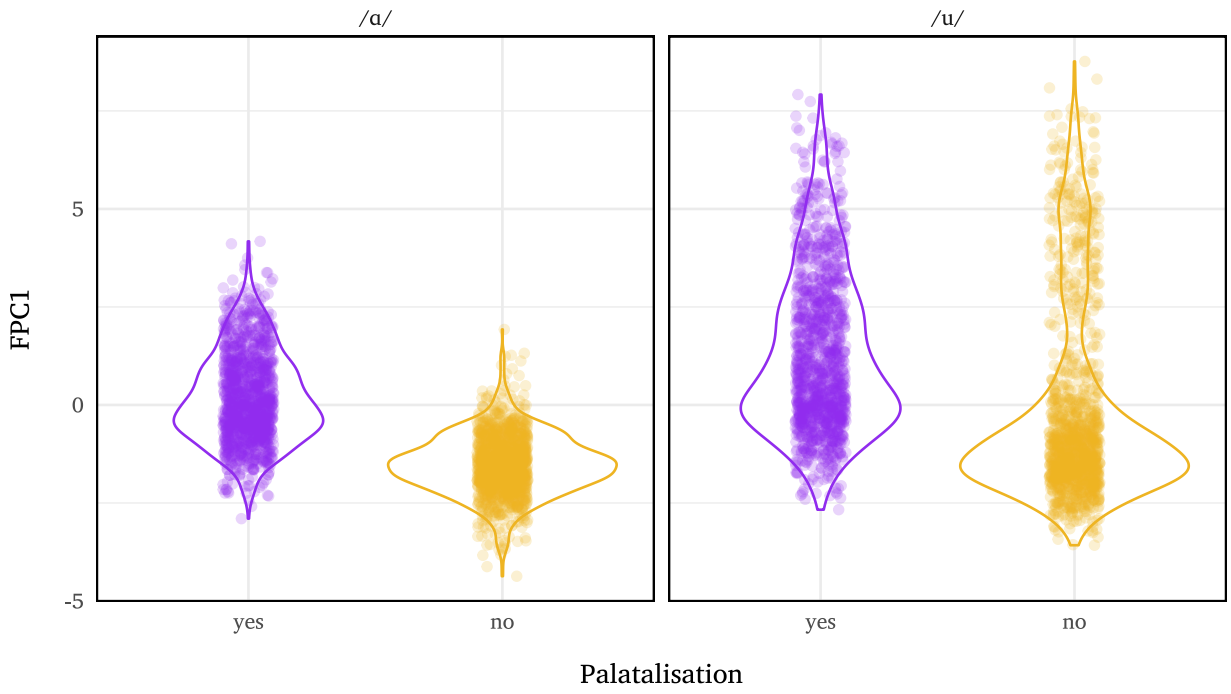


FIG. 3. FPC1 values for the different vowels.

TABLE V. FPC1 modelling.

Full model intercept	$\hat{\beta}$	SE($\hat{\beta}$)	t	$p(t)$
	-0.3	0.1	-3.79	<.001
Fixed effects		df	χ^2	$p(\chi^2)$
Age group		12	191.06	<.001
Palatalisation		9	346.08	<.001
Vowel		2	28.11	<.001
DDK		8	38.62	<.001
DDK : Age group		7	37.54	<.001
Palatalisation : Vowel		1	1.33	.248
Palatalisation : Age group		3	132.72	<.001
Palatalisation : Age group : DDK		4	31.81	<.001

418 The FPC1 model indicates that there are significant main effects for Age group, Palatal-
419 isation, Vowel and DDK. Also, Age group, Palatalisation, and DDK are significant in in-
420 teractions with each other. Vowel is significant as a main effect but not in the interaction
421 with Palatalisation. The significant differences for Vowel, but lack of significant interaction
422 between Vowel and Palatalisation indicate that FPC1 values are different for /ɑ/ and /u/
423 vowels, and are consistently different preceding palatalised and non-palatalised consonants
424 (. Figure 3). The figure shows that FPC1 is higher in /ɑ/ context, which corresponds to
425 higher F2 and a more fronter place of articulation /ɑ/ compared to /u/ in Estonian (see
426 Figure 1 lower panel in Asu and Teras, 2009, 369).

TABLE VI. FPC1 pairwise comparisons for Palatalisation and Age group.

Age group	$\hat{\beta}$	SE($\hat{\beta}$)	z	$p(z)$
3–4 years	-1.55	0.27	-5.85	<.001
6–7 years	-0.93	0.17	-5.50	<.001
10–11 years	-1.25	0.13	-9.82	<.001
Adults	-2.37	0.13	-17.95	<.001

427 We conducted pairwise comparisons for the interaction of Palatalisation and Age group
428 (Table VI), which indicates that all the age groups make a contrast between palatalised
429 and non-palatalised consonants. The relevant model estimates are shown in Figure 4. The
430 model also shows that the magnitude and consistency of the contrast made increase with age.
431 From the figure we see that for the youngest (3–4 years), the FPC1 values for palatalised
432 and non-palatalised productions show a higher difference than for the 6–7 year old group,

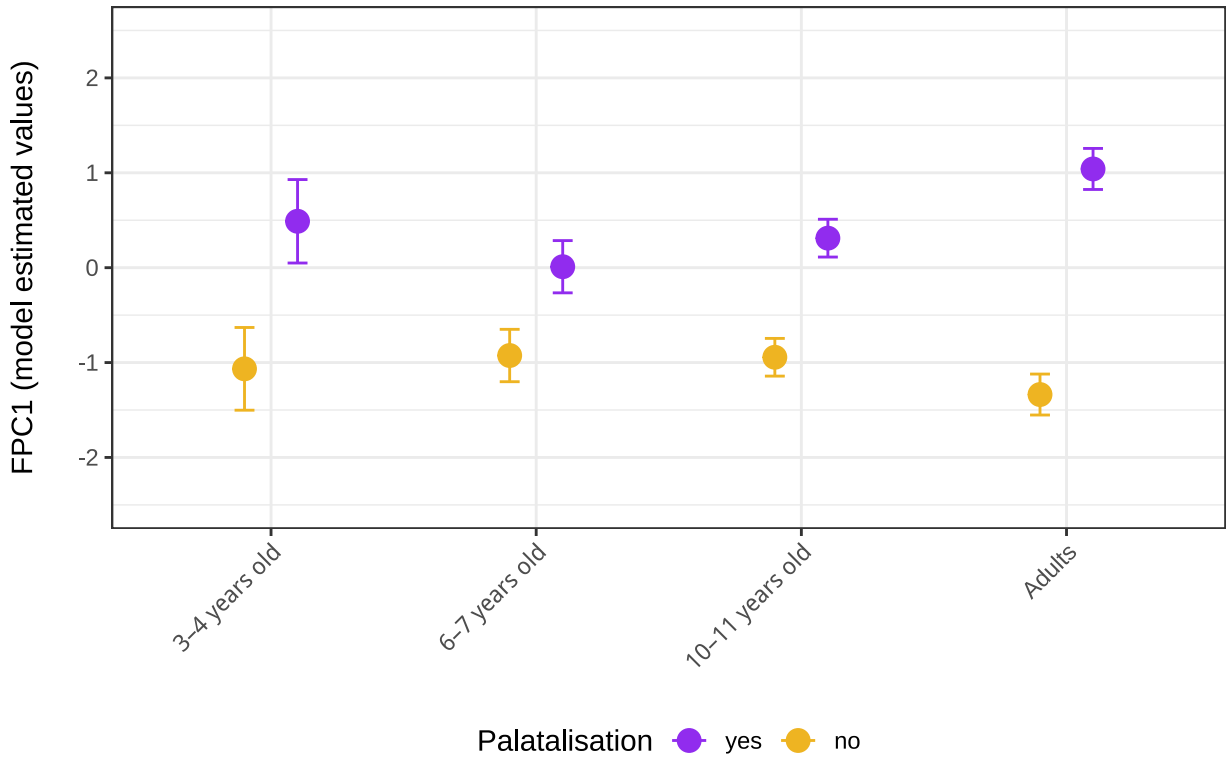


FIG. 4. Model estimates of the interaction between Age group and Palatalisation. Error bars show the 95% confidence intervals of the estimated values.

but with less consistency. After this, the magnitude of the contrast starts gradually getting bigger.

In order to interpret the role of DDK in the contrast for Palatalisation by Age group, we plotted the model estimates from three-way interaction in Figure 5. The figure indicates that there is a relationship between DDK scores and palatalisation for the youngest children and adults in which age groups with higher DDK rates also made a bigger difference between the phonological contrast (Table VII). To contextualise the DDK data in this figure, we overlaid the mean DDK syllables per second values onto these model estimates. This highlights that

441 the 3–4 year olds and 6–7 year olds have low DDK scores, but DDK scores from the 10–11
442 year olds and Adults occupy a greater range.

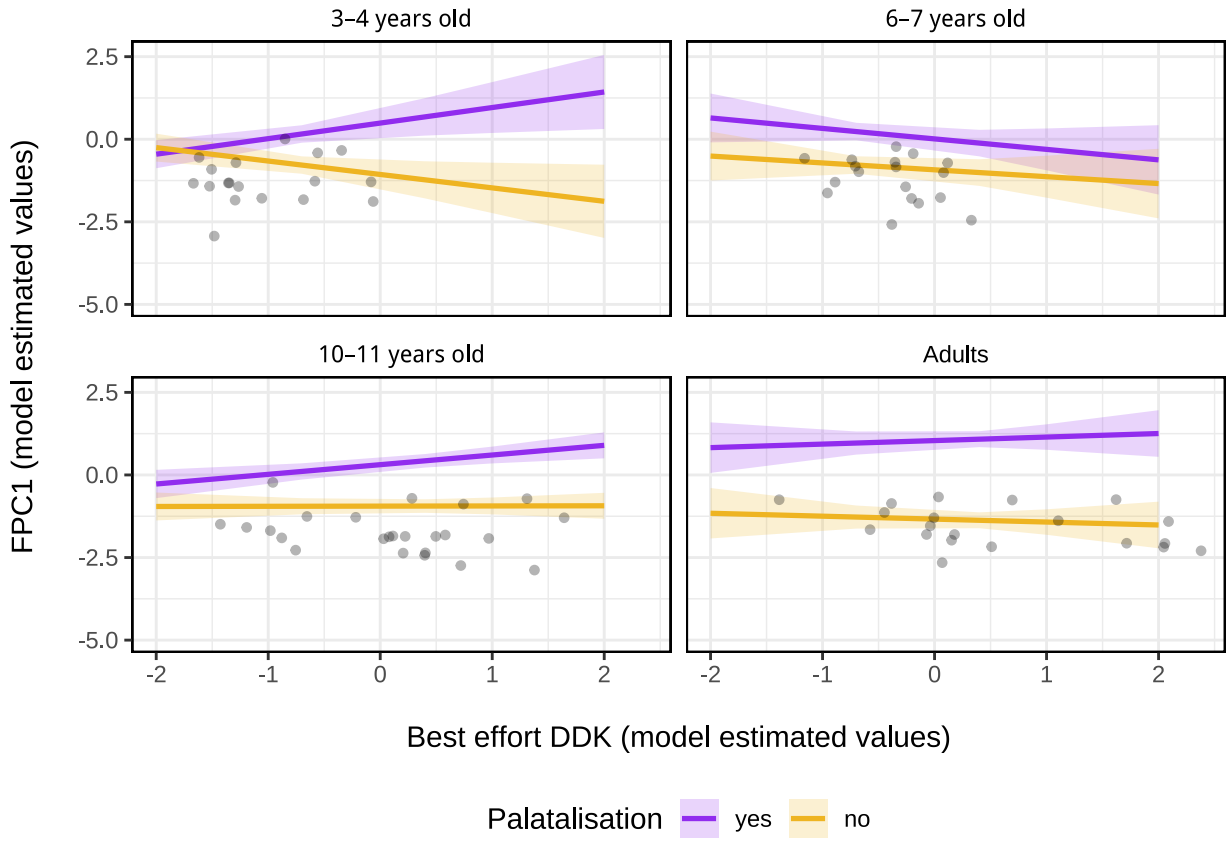


FIG. 5. Model estimates of the interaction between Age group, Palatalisation and DDK with overlaid mean DDK values and 95% confidence intervals shown on the estimate values.

TABLE VII. FPC1 pairwise comparisons for Palatalisation, Age group and DDK.

Age group	$\hat{\beta}$	SE($\hat{\beta}$)	z	$p(z)$
3–4 years	-0.65	0.21	-2.72	.001
6–7 years	0.42	0.24	1.74	.082
10–11 years	0.03	0.09	0.28	.777
Adults	-0.31	0.07	-4.62	<.001

IV. DISCUSSION

We first discuss the acquisition of oral motor control in Estonian by age (Section IV A), then the acquisition of the acoustic targets for palatalisation (Section IV B), and finally the interaction of anatomical factors and oral motor control skills in the acquisition of complex consonants (Section IV C).

A. Oral motor control in Estonian

To answer the first research question whether there are age-related differences in the oral motor control, we used a diadochokinetic (DDK) speech task, which is widely used by speech and language therapists and researchers alongside other metrics to assess the speech motor coordination of typically and atypically developing children and adults (Chu *et al.*, 2021a; Ong *et al.*, 2024). We hypothesised, based on Ha (2023); Murali *et al.* (2024), that child oral motor control skills would increase with age. This hypothesis was confirmed. We found that the number of repetitions (measured in syllables per second) of different syllables

produced by children and adults in our experiment was directly linked to their age. Overall, the youngest speakers produced the fewest number of repetitions, and the oldest speakers produced the highest. This is in line with the results of the scoping review in Kent *et al.* (2022), which analysed many studies across different ages and language backgrounds. In line with research by Devadiga and Bhat (2012), we also found that there was a reduction in the repetition rates when the place of articulation was more posterior. There were significantly fewer repetitions with /kɑ/ than /pɑ/ or /tɑ/. This can be explained by the need to move a larger mass in velar consonants compared to bilabials or apicals.

B. Acquisition of secondary palatalisation

For the second research question, we asked whether there are any age-related differences in the acoustic realisation of secondary palatalisation. We hypothesised that consonants requiring multiple articulations or very precise articulator placement are among the last sounds to be acquired (McLeod and Crowe, 2018). Cross-linguistic surveys largely support earlier work predicting a hierarchical acquisition trajectory based on articulatory complexity (Shriberg, 1993). Later acquisition of complex consonants is thought to be due to the time taken to acquire auditory and somatosensory mappings on which the motor and muscle control gesture execution relies (Ayala *et al.*, 2023; Cheng *et al.*, 2007b; Klein *et al.*, 2013; Lin and Demuth, 2015). Specifically, through gradual increasing effectiveness of auditory and somatosensory feedback and anatomical development, children acquire better control of the superior longitudinal muscle (Abakarova *et al.*, 2022, 3295) and thus lingual differentiation (Cleland and Scobbie, 2021; Dokovova *et al.*, 2023; Gibbon, 1999) and sophisticated timing

patterns (Kpogo and Chang, 2025; Noiray *et al.*, 2018). However, due to a generally flatter palate shape in younger children (Ciusa *et al.*, 2007), and a delayed change in proportion of the hard palate compared to the tongue (Vorperian *et al.*, 2005), a child's anatomy might naturally predispose them to palatalise. Previous research has indicated an overall palatalised quality to child speech in Estonian and Russian (Vihman and Vihman, 2011; Zharkova, 2005), but no detailed acoustic analysis of children's palatalisation acquisition at different ages had yet been carried out (see initial analysis in Zharkova, 2005).

Our results show that all children in our sample produce the acoustic differences between palatalised and non-palatalised consonants (Figures 4, 5, Table VI). Figure 4 indicates that the contrast develops across childhood as the distance between non-palatalised and palatalised FPC1 values is greater, but more variable, for 3–4 year olds, narrower for 6–7 year olds and then greater again for older children. This curve is reflective of a learning pattern where due to the constant adjustments to the feedback systems the performance temporarily drops and then levels out again. Table VI shows that there is a developmental increase in consistency for producing the contrast and in the magnitude of the differences between palatalised and non-palatalised. The z scores and standard errors from these comparisons show a clear developmental trajectory of increasing acoustic distance and consistency for the contrast. Figure 4 also shows an increasing magnitude of contrast with chronological age based on the model output.

Figure 4 shows estimated values for FPC1 and demonstrates that there are developmental differences for *non*-palatalised consonants as well as palatalised i.e. FPC1 values are higher for children than those of adults. These results support previous impressions that child

speech has a palatal quality (Vihman and Vihman, 2011; Zharkova, 2005) and that palatalisation is acquired early. We suggest that due to anatomical reasons and a tendency towards greater palatal contact in child speech (Cheng *et al.*, 2007a; Ciusa *et al.*, 2007), children also have to learn to *not* palatalise in their acquisition of phonological contrast. We return to the role of anatomy *in combination* with motor control in the next section.

C. Interaction of motor control and anatomical development in complex consonants

The third research question about the relationship between motor control skills and palatalisation was designed to test an apparent contradiction in the literature on child acquisition of consonants: on the one hand, complex articulations are typically acquired late due to developing lingual motor control, but on the other hand, palatalisation is reportedly acquired earlier on due to anatomy. Figure 5 allows us to explore the relationship between the acquisition of the palatalisation contrast and oral motor control at different ages.

By interpreting the interaction between DDK, group, and palatalisation we found that there is a relationship for the youngest speakers and adults between the acquisition of oral motor control and the extent to which palatalisation is phonologically contrasted (Figure 5, Table VII). This means that children with higher oral motor control needed for lingual differentiation in complex consonants realise the contrast to a greater degree. The result for adult speakers could be explained by their attempt in performing well in the tasks that were designed to be as easy for children as possible. This, in many cases, resulted in hyperarticulation of the palatalisation contrast. The results thus reflect how the adults, as

our control group, behaved when tasked with an easy experiment with some participants being more diligent than others.

Our findings advance understanding of the acquisition of complex consonants and the roles of anatomy and motor control in cross-linguistic consonant acquisition. The results show that while an acoustic impression of palatalisation is acquired early on (Kochetov and Alderete, 2011; Zharkova, 2005), the realisation of the phonological contrast between palatalised and non-palatalised takes time to acquire in adult-like fashion and proceeds in tandem with acquisition of oral motor control in young children.

D. Limitations and future work

Our study considers only the acoustic targets for palatalisation. We anticipate that the acoustic values for child palatalised consonants are articulated differently from adult realisations and that children adapt their articulatory strategy as based on anatomy (Brunner *et al.*, 2009). Our future work will analyse ultrasound tongue imaging data from the same speakers to assess lingual differentiation and consistency of articulation directly. In addition, previous work on coarticulation indicates substantial timing differences between child and adult articulations (Noiray *et al.*, 2018; Rubertus and Noiray, 2024). Our future work will also explore the acquisition of timing of the palatalisation gesture. One limitation of our study is that we use chronological age as a proxy for predicted anatomical differences based on large-scale imaging studies such as Vorperian *et al.* (2005) or palatal studies (Ciusa *et al.*, 2007). Future work could also measure hard palate length as a more direct measurement of anatomical development. Another limitation was using a DDK task with monosyllables only.

A task with trisyllabic utterances would have given us a more complete understanding on the development of the oral motor system. As we did not want to overburden children with too many task, we opted for using a simpler DDK protocol. In the production experiment with ultrasound we used audio recordings to elicit words from children. A more naturalistic approach might have yielded different results but given our lengthy experiment protocol, elicitation by imitation was the most straightforward way to elicit reliable data.

V. CONCLUSION

Our analysis contributes to a long-standing discussion in child acquisition of phonology about the order of acquisition for different phonemes (Shriberg, 1993), and the role of physiological factors in speech production acquisition (Kent, 1992). Our aim was to disentangle the contribution of, on the one hand, motor control development and, on the other hand, anatomical growth and articulator change in proportion. We found that oral motor control, measured through a diadochokinetic task (DDK), increases with age and there is a relationship between greater oral motor control and realisation of the phonemic contrast for palatalisation for the youngest children. We suggest that explaining the developmental trajectory for palatalisation acquisition is possible through an interaction of motor control and physiological development: younger children are able to produce a phonetic impression of palatalisation from the age of 3 years. It is likely that the ratio of tongue to hard palate length predisposes them to produce acoustically palatal-sounding speech. But at the same time, those who are most skilled in oral motor control are best able to differentiate palatalised and non-palatalised realisations. In this way, we argue that while anatomical de-

velopment leads straightforwardly to the production of palatalised consonants, a high degree of motor control skills is also needed for adult-like realisation of the phonological contrast. In addition, we provide initial normative data for DDK in Estonian L1-speaking children and contribute a detailed acoustic account of child acquisition of a lesser-studied consonant contrast in a lesser-studied language family.

VI. AUTHOR DECLARATIONS

A. Conflict of Interest

The authors have no conflicts of interest to disclose.

B. Ethics Approval

The experiment was approved by the Research and Ethics Committee of the University of Tartu (397/T-7). We obtained written informed consent from all participants or from their parents or legal guardians.

C. Data Availability

Data and code: https://osf.io/egmdj.

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Abakarova, D., Fuchs, S., and Noiray, A. (**2022**). "Developmental Changes in Coarticulation Degree Relate to Differences in Articulatory Patterns: An Empirically Grounded Modeling Approach," Journal of Speech, Language, and Hearing Research **65**(9), 3276–3299, doi: 10.1044/2022_JSLHR-21-00212.

Alumäe, T., Tilk, O., and Asadullah (**2018**). "Advanced rich transcription system for Estonian speech," in *Human Language Technologies - the Baltic Perspective : Proceedings of the Eighth International Conference*, edited by K. Muischnek and K. Müürisep, IOS Press, Amsterdam, pp. 1–8.

Anwyl-Irvine, A., Dalmaijer, E. S., Hodges, N., and Evershed, J. K. (**2021**). "Realistic precision and accuracy of online experiment platforms, web browsers, and devices," Behavior Research Methods **53**(4), 1407–1425, doi: 10.3758/s13428-020-01501-5.

Asu, E. L., and Teras, P. (**2009**). "Estonian," Journal of the International Phonetic Association **39**(3), 367–372, doi: 10.1017/S002510030999017X.

Ayala, S. A., Eads, A., Kabakoff, H., Swartz, M. T., Shiller, D. M., Hill, J., Hitchcock, E. R., Preston, J. L., and McAllister, T. (**2023**). "Auditory and Somatosensory Development for Speech in Later Childhood," Journal of speech, language, and hearing research: JSLHR

600 **66**(4), 1252–1273, doi: 10.1044/2022_JSLHR-22-00496.

601 Barbier, G., Perrier, P., Payan, Y., Tiede, M. K., Gerber, S., Perkell, J. S., and Ménard, L.

602 (**2020**). "What anticipatory coarticulation in children tells us about speech motor control

603 maturity," PLOS ONE **15**(4), e0231484, https://dx.plos.org/10.1371/journal.pone.

604 0231484, doi: 10.1371/journal.pone.0231484.

605 Bateman, N. (**2011**). "On the typology of palatalization," Language and Linguistic Compass

606 **5**(8), 588–602.

607 Bennett, R., Padgett, J., McGuire, G., Ní Chiosáin, M., and Bellik, J. (**2025**). "Gestural

608 timing and contrast: An Irish case study," in *Foundational approaches to Celtic linguis-*

609 *tics*, edited by A. Carnie, D. Ohala, D. Hunter, S. Prins, M. Hammond, and L. Irizarry

610 (Language Science Press, Berlin), pp. 283–341.

611 Bishai, R., Molenaar, A. M., and Scheerer, N. E. (**2026**). "Exploring the impact of at-

612 tention and sensory processing abilities on vocal motor control in children," The Journal

613 of the Acoustical Society of America **159**(3), 2902–2916, https://doi.org/10.1121/10.

614 0043030, doi: 10.1121/10.0043030.

615 Boersma, P., and Weenik, D. (**2025**). "Praat: doing phonetics by computer [Computer

616 program]. Version 6.4.27" http://www.praat.org/.

617 Brunner, J., Fuchs, S., and Perrier, P. (**2009**). "On the relationship between palate shape

618 and articulatory behavior," The Journal of the Acoustical Society of America **125**(6),

619 3936–3949, doi: 10.1121/1.3125313.

620 Cheng, H. Y., Murdoch, B., Goozée, J., and Scott, D. (**2007**a). "Electropalatographic as-

621 sessment of tongue-to-palate contact patterns and variability in children, adolescents, and

622 adults," Journal of Speech, Language, and Hearing Research **50**, 375–392.

623 Cheng, H. Y., Murdoch, B., Goozée, J., and Scott, D. (**2007**b). "Physiological development
624 of tongue-jaw coordination from childhood to adulthood," Journal of Speech, Language,
625 and Hearing Research **50**, 352–360.

626 Chu, S. Y., Foong, J. H., Lee, J., Ben-David, B. M., Barlow, S. M., and Hsu, C. (**2021**a).
627 "Oral diadochokinetic rates across languages: Multilingual speakers comparison," In-
628 ternational Journal of Language & Communication Disorders **56**(5), 1026–1036, doi:
629 10.1111/1460-6984.12653.

630 Chu, S. Y., Lee, J., Barlow, S. M., Ben-David, B., Lim, K. X., and Foong, J. H.
631 (**2021**b). "Oral-diadochokinetic rates among healthy Malaysian-Mandarin speakers: A
632 cross linguistic comparison," International Journal of Speech-Language Pathology **23**(4),
633 419–429, https://doi.org/10.1080/17549507.2020.1808701, doi: 10.1080/17549507.
634 2020.1808701 _eprint: https://doi.org/10.1080/17549507.2020.1808701.

635 Ciusa, V., Dimaggio, F. R., Sforza, C., and Ferrario, V. F. (**2007**). "Three-Dimensional
636 Palatal Development between 3 and 6 Years," The Angle Orthodontist **77**(4), 602–606,
637 doi: 10.2319/053106-221.

638 Cleland, J., and Scobbie, J. (**2021**). "The Dorsal Differentiation of Velar From Alveolar
639 Stops in Typically Developing Children and Children With Persistent Velar Fronting,"
640 Journal of Speech, Language, and Hearing Research **64**(6S), 2347–2362, doi: 10.1044/
641 2020_JSLHR-20-00373.

642 Coretta, S., and Sakr, G. (**2026**). "Multivariate Analyses of Tongue Contours from Ul-
643 trasound Tongue Imaging," Language and Speech 00238309261419120, doi: 10.1177/

644 00238309261419120.

645 Cychosz, M. (**2021**). "The coarticulation-duration relationship in early Quechua speech,"
646 Journal of Phonetics **87**, doi: 10.1016/j.wocn.2021.101052.

647 Davis, B., and Bedore, L. (**2013**). *An emergence approach to speech acquisition: Doing and*
648 *knowing* (Psychology Press, New York, NY).

649 Devadiga, D. N., and Bhat, J. (**2012**). "Oral Diadokokinetic Rate - An In-
650 sight into Speech Motor Control," https://www.semanticscholar.org/
651 paper/Oral-Diadokokinetic-Rate-An-Insight-into-Speech-Devadiga-Bhat/
652 957ee2e7ec4454593397c162e8668b9bc72c4602.

653 Dokovova, M., Sugden, E., Cartney, G., Schaeffler, S., and Cleland, J. (**2023**). "Tongue
654 Shape Complexity in Children With and Without Speech Sound Disorders," Journal
655 of Speech, Language, and Hearing Research **66**(7), 2164–2183, doi: 10.1044/2023_
656 JSLHR-22-00472.

657 Duffy, J. R. (**2013**). *Motor speech disorders: substrates, differential diagnosis, and manage-*
658 *ment*, third edition ed. (Elsevier, St. Louis, Missouri).

659 Fletcher, S. G. (**1989**). "Palatometric Specification of Stop, Affricate, and Sibilant Sounds,"
660 Journal of Speech, Language, and Hearing Research **32**(4), 736–748, doi: 10.1044/jshr.
661 3204.736.

662 Gadesmann, M., and Miller, N. (**2008**). "Reliability of speech diadochokinetic test measure-
663 ment," International Journal of Language & Communication Disorders **43**(1), 41–54, doi:
664 10.1080/13682820701234444.

Gafos, A. I. (**2002**). "A Grammar of Gestural Coordination," Natural Language & Linguistic Theory **20**(2), 269–337, doi: 10.1023/A:1014942312445.

Gibbon, F. E. (**1999**). "Undifferentiated Lingual Gestures in Children With Articulation/Phonological Disorders," Journal of Speech, Language, and Hearing Research **42**(2), 382–397, doi: 10.1044/jslhr.4202.382.

Gubian, M. (**2025**). "landmarkregUtils: Utilities for Landmark Registration," R package version 0.0.0.9000, commit ec6be48fe86289b6302e5ba9141b1d6f15da0a4b, https://github.com/uasolo/landmarkregUtils.

Gubian, M., Torreira, F., and Boves, L. (**2015**). "Using Functional Data Analysis for investigating multidimensional dynamic phonetic contrasts," Journal of Phonetics **49**, 16–40, doi: 10.1016/j.wocn.2014.10.001.

Guenther, F. H. (**2006**). "Cortical interactions underlying the production of speech sounds," Journal of Communication Disorders **39**(5), 350–365, https://linkinghub.elsevier.com/retrieve/pii/S002199240600058X, doi: 10.1016/j.jcomdis.2006.06.013.

Ha, S. (**2023**). "Oral diadochokinetic production in children with typical speech development and speech–sound disorders," International Journal of Language & Communication Disorders **58**(5), 1783–1798, doi: 10.1111/1460-6984.12908.

Happ, C., and Greven, S. (**2018**). "Multivariate Functional Principal Component Analysis for Data Observed on Different (Dimensional) Domains," Journal of the American Statistical Association **113**(522), 649–659, doi: 10.1080/01621459.2016.1273115.

Howson, P., and Redford, M. A. (**2021**). "The Acquisition of Articulatory Timing for Liquids: Evidence From Child and Adult Speech," Journal of Speech, Language, and Hearing

687 Research **64**(3), 734–753, doi: 10.1044/2020_JSLHR-20-00391.

688 Iskarous, K., and Kavitskaya, D. (**2018**). "Sound change and the structure of synchronic
689 variability: Phonetic and phonological factors in Slavic palatalization," Language **94**(1),
690 43–83.

691 Jolliffe, I. T., and Cadima, J. (**2016**). "Principal component analysis: a review and recent
692 developments," Philosophical Transactions of the Royal Society A: Mathematical, Physical
693 and Engineering Sciences **374**(2065), 20150202, doi: 10.1098/rsta.2015.0202.

694 Kabakoff, H., Harel, D., Tiede, M., Whalen, D. H., and McAllister, T. (**2021**). "Extending
695 Ultrasound Tongue Shape Complexity Measures to Speech Development and Disorders,"
696 Journal of Speech, Language, and Hearing Research **64**(7), 2557–2574, doi: 10.1044/
697 2021_JSLHR-20-00537.

698 Kent, R. (**1992**). "The biology of phonological development," in *Phonological development:*
699 *Models, research, implications*, edited by C. Ferguson, L. Menn, and C. Stoel-Gammon
700 (York Press, Timonium, MD), pp. 65–90.

701 Kent, R. D., Kim, Y., and Chen, L.-m. (**2022**). "Oral and Laryngeal Diadochokinesis Across
702 the Life Span: A Scoping Review of Methods, Reference Data, and Clinical Applications,"
703 Journal of Speech, Language, and Hearing Research **65**(2), 574–623, doi: 10.1044/2021_
704 JSLHR-21-00396.

705 Kidd, E., and Garcia, R. (**2022**). "How diverse is child language acquisition research?,"
706 First Language **42**(6), 703–735, doi: 10.1177/01427237211066405.

707 Klein, H. B., McAllister Byun, T., Davidson, L., and Grigos, M. I. (**2013**). "A Multi-
708 dimensional Investigation of Children's /r/ Productions: Perceptual, Ultrasound, and

709 Acoustic Measures," American Journal of Speech-Language Pathology **22**(3), 540–553,
710 doi: 10.1044/1058-0360(2013/12-0137).
711 Kochetov, A. (**2002**). "Production, perception, and emergent phonotactic patterns: A case
712 of contrastive palatalization," PhD Thesis, University of Toronto, Toronto.
713 Kochetov, A. (**2017**). "Acoustics of Russian voiceless sibilant fricatives," Journal of the
714 International Phonetic Association **47**(3), 321–348.
715 Kochetov, A., and Alderete, J. (**2011**). "Patterns and scales of expressive palatalization:
716 Experimental evidence from Japanese," Canadian Journal of Linguistics/Revue canadienne
717 de linguistique **56**(3), 345–376, doi: 10.1017/S0008413100002048.
718 Kpogo, F., and Chang, C. B. (**2025**). "Coarticulation and coordination in phonological
719 development: Insights from children's and adults' production of complex–simplex stop
720 contrasts in Gã," Journal of Phonetics **108**, 101–378, doi: 10.1016/j.wocn.2024.101378.
721 Lahtein-Kürsa, M., Padrik, M., Daniutė, S., Kairienė, D., Martikainen, A.-L., Vanhala-
722 Haukijärvi, M., and Mailend, M.-L. (**2025**). "Diagnostic Features of Childhood Apraxia
723 of Speech: A Survey Study of Estonian, Finnish, and Lithuanian Speech-Language
724 Pathologists," American Journal of Speech-Language Pathology **34**(1), 97–117, doi:
725 10.1044/2024_AJSLP-24-00035.
726 Liiv, G. (**1965**). "Preliminary Remarks on the Acoustic Cues for Palatalization in Estonian,"
727 Phonetica **13**(1-2), 59–64, doi: 10.1159/000258469.
728 Lin, S., and Demuth, K. (**2015**). "Children's acquisition of english onset and coda /l/:
729 Articulatory evidence," Journal of Speech, Language, and Hearing Research **58**(1), 13–27,
730 doi: 10.1044/2014_jslhr-s-14-0041.

731 Malmi, A., and Lippus, P. (**2019**). "Keele asend eesti palatalisatsioonis / The position of
732 the tongue in Estonian palatalization," Eesti ja Soome-Ugri Keeleteaduse Ajakiri **10**(1),
733 105–128.

734 Malmi, A., Lippus, P., and Meister, E. (**2023**). "Spectral and temporal properties of Es-
735 tonian palatalization," Journal of the International Phonetic Association **53**(3), 748–773,
736 doi: 10.1017/S0025100321000360.

737 Malmi, A., and Nance, C. (**2025**). "The production of articulatorily complex consonants in
738 Estonian by children and adults," in *Proceedings of Fonetik 2025*, Växjö, Sweden.

739 McLeod, S., and Crowe, K. (**2018**). "Children's Consonant Acquisition in 27 Languages: A
740 Cross-Linguistic Review," American Journal of Speech-Language Pathology **27**(4), 1546–
741 1571, doi: 10.1044/2018_AJSLP-17-0100.

742 Meister, E., and Meister, L. (**2021**). "Developmental Changes of Vowel Acoustics in
743 Adolescents," in *Interspeech 2021*, ISCA, pp. 4014–4018, doi: 10.21437/Interspeech.
744 2021-1649.

745 Mendes Tavares, E. L., Brasolotto, A. G., Rodrigues, S. A., Benito Pessin, A. B., and
746 Garcia Martins, R. H. (**2012**). "Maximum Phonation Time and s/z Ratio in a Large Child
747 Cohort," Journal of Voice **26**(5), 675.e1–675.e4, doi: 10.1016/j.jvoice.2012.03.001.

748 Murali, M., Cleland, J., Young, D., Stuart-Smith, J., Taylor, L., Smith, A., and Kuschmann,
749 A. (**2024**). "Comparing DDK rate variability in typically developing children in central
750 Scotland," San Diego, CA.

751 Nance, C., and Kirkham, S. (**2020**). "The acoustics of three-way lateral and nasal palatali-
752 sation contrasts in Scottish Gaelic," Journal of the Acoustical Society of America **147**(4),

753 2858–2872.

754 Noiray, A., Abakarova, D., Rubertus, E., Krüger, S., and Tiede, M. (**2018**). "How Do
755 Children Organize Their Speech in the First Years of Life? Insight From Ultrasound
756 Imaging," Journal of Speech, Language, and Hearing Research **61**(6), 1355–1368, doi:
757 10.1044/2018_JSLHR-S-17-0148.

758 Ong, Y. Q., Lee, J., Chu, S. Y., Chai, S. C., Gan, K. B., Ibrahim, N. M., and Barlow,
759 S. M. (**2024**). "Oral-diadochokinesis between Parkinson's disease and neurotypical elderly
760 among Malaysian-Malay speakers," International Journal of Language & Communication
761 Disorders **59**(5), 1701–1714, doi: 10.1111/1460-6984.13025.

762 R Core Team (**2025**). *R: A Language and Environment for Statistical Computing* (R Foun-
763 dation for Statistical Computing, Vienna, Austria), www.R-project.org.

764 Ramsay, J. (**2003**). "fda: Functional Data Analysis" doi: 10.32614/CRAN.package.fda,
765 institution: Comprehensive R Archive Network Pages: 6.3.0.

766 Rubertus, E., and Noiray, A. (**2018**). "On the development of gestural organization: A
767 cross-sectional study of vowel-to-vowel anticipatory coarticulation," PLOS ONE **13**(9),
768 1–21, doi: 10.1371/journal.pone.0203562.

769 Rubertus, E., and Noiray, A. (**2024**). "Children's coarticulation patterns as a window to
770 the phonology-phonetics interface," in *Proceedings of the International Seminar on Speech*
771 *Production*, edited by C. Fougeron and P. Perrier, Autrans, France.

772 Shaw, J., Sejin, O., Durvasula, K., and Kochetov, A. (**2021**). "Articulatory coordination
773 distinguishes complex segments from segment sequences," Phonology **38**, 437–477, doi:
774 10.1017/S0952675721000269.

775 Shriberg, L. D. (**1993**). "Four New Speech and Prosody-Voice Measures for Genetics Re-
776 search and Other Studies in Developmental Phonological Disorders," Journal of Speech,
777 Language, and Hearing Research **36**(1), 105–140, doi: 10.1044/jshr.3601.105.

778 Spreafico, L., Pucher, M., and Matosova, A. (**2018**). "UltraFit: A Speaker-friendly
779 Headset for Ultrasound Recordings in Speech Science," pp. 1517–1520, doi: 10.21437/
780 Interspeech.2018-995.

781 Stackhouse, Joy, P. W. (**2000**). "Rate, accuracy and consistency: diadochokinetic perfor-
782 mance of young, normally developing children," Clinical Linguistics & Phonetics **14**(4),
783 267–293, doi: 10.1080/02699200050023985.

784 Teras, P., and Pajusalu, K. (**2014**). "Palatalisatsioonist ja prepalatalisatsioonist spontaanses
785 eesti keeles / palatalization and prepalatalization in Estonian spontaneous speech," Keel
786 ja kirjandus **4**, 257–269.

787 Timm, L. (**1977**). "A child's acquisition of Russian phonology," Journal of Child Language
788 **4**(3), 329–339, doi: 10.1017/S0305000900001720.

789 Tremblay, S., Shiller, D. M., and Ostry, D. J. (**2003**). "Somatosensory basis of speech
790 production," Nature **423**(6942), 866–869, doi: 10.1038/nature01710.

791 Turton, D. (**2017**). "Categorical or gradient? An ultrasound investigation of /l/-darkening
792 and vocalization in varieties of English," Laboratory Phonology: Journal of the Association
793 for Laboratory Phonology **8**(1), 1–31.

794 Vihman, M. (**1971**). "On the acquisition of Estonian," Papers and Reports on Child Lan-
795 guage Development **3**, 53–94.

Vihman, M., and Vihman, V.-A. (**2011**). "From first words to segments: A case study in phonological development," in *Experience, variation and generalization: Learning a first language*, edited by I. Arnon and E. Clark (John Benjamins, Amsterdam), pp. 109–133.

Vorperian, H. K., Kent, R. D., Lindstrom, M. J., Kalina, C. M., Gentry, L. R., and Yandell, B. S. (**2005**). "Development of vocal tract length during early childhood: A magnetic resonance imaging study," The Journal of the Acoustical Society of America **117**(1), 338–350, doi: 10.1121/1.1835958.

Winter, B. (**2020**). *Statistics for Linguists: An Introduction Using R* (Routledge).

Wrench, A. (**2024**). *Articulate Assistant Advanced v221.5.11* (Articulate Instruments, Edinburgh).

Zharkova, N. (**2005**). "Strategies in the acquisition of segments and syllables in Russian-speaking children," Leiden Papers in Linguistics **2**(1), 189–213.

Zharkova, N. (**2018**). "An Ultrasound Study of the Development of Lingual Coarticulation during Childhood," Phonetica **75**(3), 245–271, doi: 10.1159/000485802.

Ziegler, W., Schölderle, T., Brendel, B., Amsellem, J., and Et., A. (**2019**). "Higher-Faster-Farther: Maximum Performance Tests in the Assessment of Neurogenic Speech Impairment," Folia Phoniatrica et Logopaedica https://sci-hub.st/https://karger.com/article/doi/10.1159/000495784, doi: 10.1159/000495784.